

INPUT: Raw GenPept Sequences · Bacteria · Archaea · Fungi

Data Retrieval & Taxonomic Scoping

Source: GenPept / NCBI Protein (CDS translations from GenBank)

Scope: Bacteria (taxid 2) · Archaea (taxid 2157) · Fungi (taxid 4751)

Window: 01 Apr 2025 31 Jul 2025

Quality Control

Length: 10 L 200 aa (both bounds inclusive)

Composition: standard 20 aa only - remove U, O, B, Z, J, X

Annotation: exclude fragment · hypothetical · putative · predicted · LOW QUALITY

Precursor Handling

Retain if 10 L 200 aa; mature region assumed within range

Flag in metadata excluded from main dataset (n = 287 records)

Sensitivity analysis: reintroducing n = 287 no statistically significant change

IPG-Based Deduplication

Query each record against NCBI Identical Protein Groups (IPG) via Entrez
IPG clusters 100%-identical AA sequences: GenBank · RefSeq · SwissProt · PDB
One representative per cluster (most complete DEFINITION / lex. smallest accession)

Annotation-Based Labeling

Step 1: False-positive filter exclude ampC / ampD / ampR / ampS

Step 2: Primary annotation scan DEFINITION field for activity-indicating phrases

Step 3: CDS product check additionally scan CDS product field (if available)

Step 4: Assign AMP (antimicrobial · bacteriocin · lantibiotic) or non-AMP

Homology-Controlled Partitioning

CD-HIT clustering at 90% pairwise sequence identity

Cluster-intact train/validation/test partitioning

Post-split MMseqs2 verification

OUTPUT: GenPept-Curated-2025 (n = 11,000)

5,500 AMP · 5,500 non-AMP · 10 - 200 aa · Bacteria · Archaea · Fungi